MIL-I-60166(MU)
14 May 1964

MILITARY SPECIFICATION

INVERTER, POWER, STATIC: 10516412

1. SCOPE

1.1 This specification covers one type of static power inverter, that is completely transistorized. The inverter is capable of inverting an 18 to 30 volt direct current potential to a 115 volt, 400 cycles per second, alternating current potential, delivering at full load 150 volt amperes.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Military

MIL-F-13926	Fire Control Materiel: General Specification Governing the Manufacture and Inspection of
MIL-P-14232	Parts, Equipment and Tools for Ordnance Materiel, Packaging of
MIL-I-26600	Interference Control Requirements, Aeronautical Equipment
MIL-I-45607	Inspection Equipment, Supply and Maintenance for Ordnance

STANDARDS

Federal

FED-STD-595	Colors

FSC 1220

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Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	Quality Assurance Terms and Definitions
MIL-STD-171	Finishing of Metal and Wood Surfaces
MIL-STD-202	Test Methods for Electronic and Electrical Component Parts

DRAWINGS

U.S. Army Munitions Command

D10516412	Inverter, Power, Static: 10516412

PACKAGING DATA SHEET

MIL-P-14232/P10516412	Packaging of Inverter, Power, Static: 10516412

(Copies of specifications, standards, and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Qualification. - The Inverter, Power, Static: 10516412 furnished under this specification shall be a product which has been tested and passed the qualification tests specified herein, and has been listed on or approved for listing on the applicable qualified products list (see 6.3).

3.2 Fabrication. - The inverter shall be completely transistorized and manufactured in accordance with Drawing D10516412 and drawings pertaining thereto.

3.2.1 Inclosure. - The inclosure shall be made of metal capable of withstanding all the requirements of this specification. The inclosure shall be finished in accordance with Standard MIL-STD-171, System 20.8. The color shall be lusterless black, No. 37038 as specified in Standard Fed-STD-595. The dried finish shall show no running or color separation and shall present a smooth lusterless finish with no evidence of bare spots.

3.2.2 Circuit components. - The circuitry shall employ standard commercially available components wherever practical. Special components, if used, shall be capable of fabrication by mass production methods.

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3.2.3 Circuitry. - An electrical schematic diagram shall be provided with each inverter. This schematic shall provide all the information required to trouble shoot and repair the inverter. If modules are used the circuitry shall be shown.

3.3 General specification. - The inverter shall meet the requirements of Specification MIL-F-13926, as specified below. Should any conflict exist between the general specification and this specification, this specification shall take precedence.

(a) Order of precedence
(b) Dimensions and tolerances
(c) Inorganic protective surface finishes
(d) Part identification and marking
(e) Workmanship

3.4 Reliability. - The inverter shall be operated continuously for 600 hours in a horizontal position and 400 hours in a position at 180° from the horizontal with a direct current (dc) input of 30 volts (v) and at an ambient temperature of 150 ±4°F. The rated full load output shall be maintained at 150 volt-amperes (VA) with the least efficient power factor between 0.8 leading and 0.6 lagging. Subsequent to this operation, the inverter shall meet the requirements of 3.7 at room ambient temperature (75 ± 15°F).

3.5 Environmental.

3.5.1 Temperature shock. - The inverter shall show no evidence of physical damage and shall meet the requirements of 3.7 at room ambient temperature after exposure to the following temperature excursions in uninterruped sequence. Unless otherwise specified ambient temperature extremes shall be within ±4°F.

(a) A gradual increase from room ambient temperature with a 2 hour exposure at 160°F (see 6.4).
(b) Decrease (60 ± 1 minutes) to 65°F and maintained 2 hours
(c) Gradual reduction in temperature with a 2 hour exposure at -80°F (see 6.4).
(d) Gradual increase in temperature to 75 ± 15°F.

3.5.2 Vibration. - The inverter shall show no evidence of damage when vibrated for 10 minutes in each direction of the 3 mutually perpendicular axes (total of 30 minutes) relative to its normal operating position. The vibration shall be a simple harmonic motion having a 0.03 inch amplitude (0.06 inch total excursion). The frequency shall vary uniformly over a range of 10 to 55 cycles per second (cps) and return to 10 cps in 1 minute ± 15 seconds. During vibration the inverter shall meet the requirements of 3.7.

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3.5.3 Shock. - The inverter shall show no evidence of physical damage and shall meet the requirements of 3.7 after being exposed to shock impulses of 25 gravitational units (G) each. Pulse duration of each shock impulse shall be 11 ± 1 milliseconds. With the exception of the receptacle side, 4 shock impulses shall be applied perpendicular to each side.

3.5.4 Salt spray. - The inverter shall show no evidence of finish deterioration and shall meet the requirements of 3.7 after exposure to the salt spray conditions below:

a. The salt-solution concentration shall be 5% with a specific gravity between 1.0268 and 1.0413.

b. Solution shall have a pH value between 6.5 and 7.2 at 95 ± 3.6°F.

c. Nozzle pressure of 15 ± 3 lb per square inch with an orifice of from 0.02 to 0.03 inch diameter.

3.5.5 Humidity. - The inverter shall meet the requirements of 3.7 after being subjected to the following relative humidity (RH) conditions:

a. Water shall have a pH value between 6.5 and 7.5 at 77°F.

b. Six hours exposure at 95% RH; 4½ hours shall be at an ambient temperature of 115°F.

c. Temperature variation of ± 9°F once each hour during the next 6 hours at an ambient temperature of 80°F.

d. Relative humidity shall be raised to 100% with condensation once each hour during the last 6 hours. (Duration of 100% RH with condensation optional and need not coincide with hourly temperature variation.)

3.5.6 Rain. - The inverter shall be capable of withstanding a rainfall of 4 inches per hour for a period of 2 hours, without visible damage or defect, and shall subsequently meet the requirements of 3.7.

3.6 Electrical.

3.6.1 Protection.

3.6.1.1 Low input voltage protection. - The inverter shall not be damaged when subjected to continuous input voltages below 18 vdc with the full load connected to the output. Any output power available below 18 vdc input shall be 115 vac ± 10% at 400 cps ± 10%.

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3.6.1.2 Reverse polarity protection. - The inverter shall be inherently protected against applied input voltage of reverse polarity.

3.6.1.3 Short circuit and overload protection. - The inverter shall utilize automatic reset self-protective methods against short circuits and overloads in excess of 150% of the full load (150VA). Fuses shall not be used.

3.6.2 Isolation. - The inverter housing shall be at ground potential.

3.6.3 Insulation resistance. - Unless otherwise specified on applicable drawings, all insulation between electrically isolated circuits shall have a minimum resistance of 100 megohms with 500 vdc applied.

3.6.4 Dielectric strength. - There shall be no puncture of insulation, no sparkover, or decrease in dielectric strength when insulated areas, that are electrically isolated from ground and from each other, are subjected to an ac root mean square (rms) potential of 1000 v plus twice the rated voltage at 60 cps for a period of not less than 3 seconds nor more than 5 seconds.

3.7 Performance. - The inverter shall meet the requirements of 3.7.1 thru 3.7.6 without spasmodic or intermittent operation with an input voltage of 24 ± 6 vdc. The voltage shall be applied to input terminals D and E, which shall be positive with respect to terminal B. All terminal designations are those shown on Drawing D10516412.

3.7.1 Input voltage surge. - The inverter shall show no evidence of physical or electrical damage and shall exhibit no increase in output voltage after being subjected to five dc voltage spikes of 80 volts each for 125 ± 25 milliseconds at the rate of 5 spikes per minute.

3.7.2 Output voltage. - The inverter shall have a regulated nominal ac output voltage of 115 v rms.

3.7.3 Output voltage regulation. - The inverter output voltage regulation shall be ± 5% for any combination of load variation from no load to full load (150VA).

3.7.4 Output frequency. - The inverter output sinusoidal wave frequency shall be 400 cps ± 1%.

3.7.5 Harmonic distortion. - The inverter output shall be free of electrical voltage spikes and shall have a harmonic distortion of less than 4% for all loads and all power factor ranges between 0.8 leading and 0.6 lagging. With a constant nominal unity power factor and at a nominal load of 75 VA the maximum variation of harmonic distortion shall be 1.5%.

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3.7.6 Power factor range. - The inverter shall be capable of operating with power factor loads ranging from 0.8 leading and 0.6 lagging.

3.8 Operability. - The inverter shall be thermally stabilized at an ambient temperature of 150 ± 4°F and operated continously without additional heat sinks for 8 hours with a full load of 150 VA at the least efficient power factor between 0.8 leading and 0.6 lagging. Immediately following the 8 hour operation the inverter shall meet the requirements of 3.7. Subsequent to meeting the high temperature requirement, the inverter shall be thermally stabilized at -65 ± 4°F and again operated at aforementioned full load and power factor for 1 hour (minimum). Immediately following the 1 hour operation, the inverter shall meet the requirements of 3.7.

3.9 Efficiency. - After operating for a maximum period of 10 minutes the inverter shall operate continuously with a minimum efficiency of 60% under a full load at 150°F with a dc input of 24v.

3.10 Radio interference. - The inverter shall meet the requirements of Specification MIL-I-26600 (Class III) for radio interference suppression.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. - Unless otherwise specified in the contract or purchase order the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 General provisions. - The quality assurance provisions of this specification and of other documents referenced herein form the basis for inspection to be performed by the supplier. Inspection of components and subassemblies not covered by Specifications or Supplementary Quality Assurance Provisions (SQAP) shall be in accordance with 4.3.2. Definitions of inspection terms not otherwise defined herein shall be as listed in Standard MIL-STD-109.

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4.2 Classification of tests. - Inspection and testing shall be classified as follows:

1 - Qualification tests
2 - Quality conformance tests

4.2.1 Qualification tests. - Qualification inspection shall consist of the examinations and tests specified for all the requirements of this specification to determine compliance with 3.1. These tests shall be performed in a Government approved laboratory, or by a Government representative at the manufacturer's facility.

4.2.1.1 Specimens for test. - Three specimens shall be submitted for qualification approval. The specimens shall be finished products representative of the materials and workmanship to be applied in normal production by the manufacturer.

4.2.1.2 Qualification approval. - Facilities submitting specimens for Qualified Products List (QPL) shall be placed on the QPL if the specimens submitted meet all of the conditions of Table I of this specification. Failure to meet all requirements shall be cause for refusal to grant qualification approval.

TABLE I - QUALIFICATION TESTS

CHARACTERISTIC	REQ'T PAR.	TEST PAR.
Fabrication	3.2	4.6.1
Inclosure	3.2.1	4.6.2
Circuit components	3.2.2	4.6.3
Circuitry	3.2.3	4.6.4
General specifications	3.3	MIL-F-13926 - Visual
Reliability	3.4	4.6.5
Temperature shock	3.5.1	4.6.6.1
Vibration	3.5.2	4.6.6.2
Shock (mechanical)	3.5.3	4.6.6.3
Salt spray	3.5.4	4.6.6.4
Humidity	3.5.5	4.6.6.5
Rain	3.5.6	4.6.6.6
Low input voltage protection	3.6.1.1	4.6.7.1
Reverse polarity protection	3.6.1.2	4.6.7.2
Short circuit and overload protection	3.6.1.3	4.6.7.3
Isolation	3.6.2	4.6.7.4
Insulation resistance	3.6.3	4.6.7.5
Dielectric strength	3.6.4	4.6.7.6
Performance	3.7	4.6.8
Input voltage surge	3.7.1	4.6.8.1

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TABLE I - QUALIFICATION TESTS (Cont)

CHARACTERISTIC	REQ'T PAR.	TEST PAR.
Output voltage	3.7.2	4.6.8.2
Output voltage regulation	3.7.3	4.6.8.3
Output frequency	3.7.4	4.6.8.4
Harmonic distortion	3.7.5	4.6.8.5
Power factor	3.7.6	4.6.8.6
Operability	3.8	4.6.9
Efficiency	3.9	4.6.10
Radio interference	3.10	4.6.11

4.2.2 Quality conformance tests. - Quality conformance tests shall consist of an inspection for those characteristics listed in Table II and control tests under 4.4. Unless otherwise specified, the finished product shall be tested to determine compliance with the applicable requirements.

4.2.2.1 Final acceptance. - Final acceptance of a lot of finished inverters shall be withheld by the government inspector until the sample products, representative of that lot, have satisfactorily met the applicable requirements of this specification.

4.3 Inspection provisions.

4.3.1 Inspection lot. - Unless otherwise specified by the contracting officer, inspection lot size, formation, and presentation of lots shall be in accordance with "Submission of Product" as specified in Standard MIL-STD-105.

4.3.2 Examination and tests. - Components and subassemblies shall be inspected in accordance with the inspection provisions contained in Specification MIL-F-13926. Examinations and tests related to Section 3 herein shall be performed on a defect (individual characteristic) basis in accordance with Standard MIL-STD-105, and the inspection level and sampling plans specified in Table II titled "Classification of Defects". Examinations and tests for Packaging, Packing and Marking shall be in accordance with Specification MIL-P-14232 and Section 5 herein. The tabulated classification of defects shall constitute the minimum inspection to be performed by the supplier prior to Government acceptance of rejection by lot. The Government reserves the right to inspect for any applicable requirement, and to reject individual non-conforming items.

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TABLE II - CLASSIFICATION OF DEFECTS

Use Inspection Level II of Table I with
Sampling Plan Table II-A of Standard MIL-STD-105

		REQUIREMENT	TEST PROCEDURE
CRITICAL: NONE			
MAJOR: AQL 0.65% Defective			
101.	Vibration	3.5.2	4.6.6.2
102.	Low input voltage protection	3.6.1.1	4.6.7.1
103.	Reverse polarity protection	3.6.1.2	4.6.7.2
104.	Short circuit and overload protection	3.6.1.3	4.6.7.3
105.	Isolation	3.6.2	4.6.7.4
106.	Insulation resistance	3.6.3	4.6.7.5
107.	Dielectric strength	3.6.4	4.6.7.6
108.	Performance	3.7	4.6.8
109.	Input voltage surge	3.7.1	4.6.8.1
110.	Output voltage	3.7.2	4.6.8.2
111.	Output voltage regulation	3.7.3	4.6.8.3
112.	Output frequency	3.7.4	4.6.8.4
113.	Harmonic distortion	3.7.5	4.6.8.5
114.	Power factor	3.7.6	4.6.8.6
115.	Radio interference	3.10	4.6.11
MINOR: NONE			

4.3.3 Disposition of non-conforming product. -Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and re-submittance of rejected lots shall be in accordance with "Acceptance and Rejection" as specified in Standard MIL-STD-105.

4.4 Control tests.

4.4.1 General. - One inverter shall be selected at random as a control sample from each 50 produced, or from each month's production, whichever occurs first. The control sample shall have successfully met all other acceptance tests specified herein and shall then meet the following requirements and tests:

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CONTROL TEST	REQUIREMENT	TEST PROCEDURES
Fabrication	3.2	4.6.1
Inclosure	3.2.1	4.6.2
Circuit components	3.2.2	4.6.3
Circuitry	3.2.3	4.6.4
General specifications	3.3	Visual, MIL-F-13926

4.4.2 Environmental and functional. - Three inverters shall be selected at random from each 100 produced or from each month's production, whichever occurs first. The control samples shall have successfully met all other acceptance tests specified herein, and shall then meet the following requirements and tests:

CONTROL TEST	REQUIREMENT	TEST PROCEDURE
Temperature shock	3.5.1	4.6.6.1
Shock (mechanical)	3.5.3	4.6.6.3
Salt spray	3.5.4	4.6.6.4
Humidity	3.5.5	4.6.6.5
Rain	3.5.6	4.6.6.6
Operability	3.8	4.6.9
Efficiency	3.9	4.6.10

4.4.3 Reliability. - One inverter shall be selected from each 300 produced to determine compliance with 3.4 and 4.6.5. This shall be in addition to the first inverter produced after initiation of production.

4.4.4 Control test failure. - Should any item of a control test sample fail to meet the specified test requirements, acceptance of the product will be suspended by the Government until necessary corrections have been made by the contractor and the resubmitted samples have been approved.

4.5 Inspection equipment. - Except as otherwise provided for by the contract, the contractor shall maintain all required measuring and testing equipment in accordance with the applicable requirements specified in Specification MIL-I-45607. The Government reserves the right to to use test equipment for its own independent inspections to the extent that such use will not unduly interfere with the contractor's delivery schedule.

4.5.1 Government furnished inspection equipment. - Where the contract provides for Government furnished test equipment, care and maintenance of test equipment shall be in accordance with the applicable requirements specified in Specification MIL-I-45607.

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4.5.2 Contractor furnished inspection equipment.

4.5.2.1 Contractor design. - The contractor shall design and supply inspection equipment compatible with the "Test Methods and Procedures" specified in 4.6 and with the "Component Inspection" requirements of Specification MIL-F-13926. Concept, construction, materials, dimensions, and tolerances used in design of test equipment shall be so selected and controlled as to insure that the test equipment will permit positive rejection of a product which exceeds the prescribed tolerance limit, and will reliably indicate acceptability of a product which does not exceed 90% of the prescribed tolerance limit. Construction shall be such as to facilitate routine calibration of the equipment.

4.5.2.2 Equipment used for environmental testing. - All environmental testing equipment used to determine compliance with environmental requirements (Listed below) shall conform to the "Test Facilities" requirements outlined in Specification MIL-F-13926.

REQUIREMENT	PARAGRAPH
(a) Vibration	3.5.2
(b) Temperature	3.4; 3.5.1; 3.8; 3.9
(c) Shock	3.5.3
(d) Humidity	3.5.5
(e) Rain	3.5.6

4.6 Test methods and procedures.

4.6.1 Fabrication. - The power inverter shall be subjected to a visual, tactile and dimensional inspection to determine compliance with the applicable drawings and requirements of this specification. Dimensional inspection shall be performed using standard measuring equipment.

4.6.2 Inclosure. - The material and finishes used in manufacture of the inclosure shall be examined to insure compliance with 3.2.1. In addition to visual examination, data shall be available showing evidence of material analysis prior to acceptance by certification. The external surfaces of the inverter shall be visually inspected for completeness of the specified finish and shall show no evidence of running or color separation, damage, or bare spots as specified in 3.2.1.

4.6.3 Circuit components. - The components used shall be visually examined to insure compliance with 3.2.2. In addition to visual examination, data shall be available showing evidence of acceptable component manufacture in accordance with established requirements prior to acceptance by dertification.

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4.6.4 Circuitry. - The inverter shall be visually examined to insure inclusion of the schematic in compliance with the requirements of 3.2.3. The schematic shall be visually examined for true representation of the circuitry and applicable drawings, including modules when applicable.

4.6.5 Reliability. - The inverter shall be properly positioned, secured, and operated as specified in 3.4 to determine compliance with requirements therein. Standard testing and measuring equipment shall be utilized. Subsequent to this test the inverter shall be subjected to the tests specified in 4.6.8 to 4.6.8.6 inclusive and shall meet the requirements referenced therein.

4.6.6 Environmental.

4.6.6.1 Temperature shock. - This test shall be conducted in accordance with the temperatures specified in 3.5.1. Tests and performance measurements during temperature cycling are not required for final acceptance. Upon completion of the specified temperature cycling, and return to standard ambient temperature (60° to 90°F), the inverter shall be inspected for evidence of physical damage. It shall then be subjected to the tests specified in 4.6.8 to 4.6.8.6 inclusive to determine conformance with the requirements referenced therein.

4.6.6.2 Vibration. - This test shall be conducted in accordance with the motion and amplitude specified in 3.5.2 shall be applied for a period of 10 minutes in each of the three mutually perpendicular directions (total of 30 minutes). During vibration the inverter shall be subjected to the tests specified in 4.6.8 to 4.6.8.6 inclusive and shall meet the requirements specified therein. Subsequent to vibration the inverter shall be subjected to visual and tactile examination and shall show no evidence of physical failure.

4.6.6.3 Shock. - This test shall be conducted in accordance with Procedure I of the shock test in Specification MIL-F-13926 and the conditions specified in 3.5.3. Subsequent to the shock test the inverter shall be subjected to a visual and tactile examination and shall show no evidence of physical failure. The inverter shall then be subjected to the tests specified in 4.6.8 to 4.6.8.6 and shall conform to the requirements referenced therein.

4.6.6.4 Salt spray. - This test shall be conducted in accordance with Method 101, Condition B, of Standard MIL-STD-202. Subsequent to the salt spray exposure the inverter shall be subjected to a visual examination and shall show no evidence of finish deterioration in compliance with the requirements of 3.5.7. The inverter shall then be subjected to the tests specified in 4.6.8 to 4.6.8.6 inclusive and shall meet the requirements referenced therein.

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4.6.6.5 Humidity. - This test shall be conducted in accordance with the requirements specified in 3.5.5. Upon completion of this test the cover shall be removed and the inverter shall show no evidence of damage or moisture. The inverter shall then be subjected to the tests specified in 4.6.8 to 4.6.8.6 inclusive to determine compliance with 3.5.5.

4.6.6.6 Rain. - This test shall be conducted in accordance with 3.5.6 and Procedure I of the Rain Test in Specification MIL-F-13926. Upon completion of this test the inverter shall be visually examined for evidence of water penetration. Subsequent to this test, the inverter shall be subjected to the tests specified in 4.6.8 to 4.6.8.6 inclusive to determine compliance with the requirements referenced therein.

4.6.7 Electrical.

4.6.7.1 Low input voltage protection. - This test shall be conducted using a minimum simulated load of 150 VA. Energize the inverter with a positive input of 24 ± 6 vdc applied to input terminals D & E (see Dwg. D10516412) and gradually reduce this input voltage below 18 vdc. The inverter shall then be subjected to one hour of continuous operation at the lowest input voltage below 18 vdc required to sustain output power. Any output power, produced with this minimum input voltage, shall be within the tolerances specified in 3.6.1.1. Subsequent to the application of this low input voltage, the inverter shall show no evidence of damage, in compliance with the requirements of 3.6.1.1.

4.6.7.2 Reverse polarity protection. - This test shall be conducted by applying a voltage not to exceed minus 30 volts direct current to the input terminals D and E with respect to terminal B of the inverter (see Dwg. 10516412). Subsequent to the application of this voltage, the inverter shall show no evidence of damage in compliance with the requirements of 3.6.1.2.

4.6.7.3 Short circuit and overload protection. - When energized, the inverter shall be subjected to a direct short circuit and overload in the amount specified in 3.6.1.3 for a minimum of one minute. This short circuit and overload shall cause the protective mediums in the inverter to open immediately upon application of short circuit and overload and the inverter to cease to function. After removal of the short circuit and overload, the inverter shall show no evidence of damage.

4.6.7.4 Isolation. - This test shall be conducted using standard measuring equipment to check from the inverter housing to terminal B (see Dwg. 10516412) to ascertain that the inverter housing is at ground potential in compliance with 3.6.2.

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4.6.7.5 Insulation resistance. - This test shall be conducted by measuring the insulation resistance between mutually insulated parts by means of a megohm bridge or other approved method with the DC potential applied as specified in 3.6.3 to determine compliance therewith.

4.6.7.6 Dielectric strength. - The inverter shall be tested in accordance with the potential and time duration specified in 3.6.4, and the test Method 301 of Standard MIL-STD-202 to determine compliance with the requirements specified in 3.6.4.

4.6.8 Performance. - The inverter shall be observed for spasmodic or intermittent operation while being subjected to the tests of 4.6.8 to 4.6.8.6 inclusive to determine conformance with 3.7. Any evidence of spasmodic of intermittent operation during these tests shall be cause for rejection.

4.6.8.1 Input voltage surge. - While energized with its normal rated input voltage, apply the input voltage spikes specified in 3.7.1 for the required time period. The inverter shall be examined during this test to insure against increased output voltage and subsequent to the application of the voltage spikes, the inverter shall show no evidence of damage in compliance with the requirements specified in 3.7.1.

4.6.8.2 Output voltage. - This test shall be conducted utilizing standard measuring equipment. The input voltage shall be established at the minimum voltage, the mean voltage and the maximum voltage of the 18 to 30 Vdc operational range. The measured rms output voltage shall meet the requirements of 3.7.2.

4.6.8.3 Output voltage regulation. - With the inverter energized and any combination from no load (OVA) one-half load (75VA) to full load (150VA) variation applied, the input voltage shall be measured in 3 volt increments from 18 to 30 volts dc with standard measuring equipment. The measured output voltage shall be within the specified tolerance of 3.7.3.

4.6.8.4 Output frequency. - This test shall be conducted utilizing standard measuring equipment. The input voltage shall be established at the minimum voltage, the mean voltage, and the maximum voltage of the 18 to 30 Vdc operational voltage range. The measured output frequency shall be within the tolerance specified in 3.7.4.

4.6.8.5 Harmonic distortion. - Energize the inverter and establish the input voltages at the minimum voltage, the mean voltage and the maximum voltage of the 18 to 30 Vdc operational voltage range for this test. The inverter shall have a harmonic distortion of less than 4% as specified in 3.7.5 when subjected to any load combination for each of the input voltages from no load (OVA), one-half load (75VA) and full load (150VA) and the minimum loads required to obtain the power factor range of 0.8 leading and 0.6 lagging. The harmonic distortion variation shall be 1.5% or less as specified in 3.7.5 when the inverter is subjected to a nominal load of 75VA at constant unity power factor for 15 minutes at each of the input voltages. The measured harmonic distortion shall not exceed the tolerances specified and there shall be no evidence of electrical voltage spikes during test in compliance with 3.7.5. All measurements shall be made with standard measuring equipment.

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4.6.8.6 Power factor range. - This test shall be conducted utilizing standard measuring equipment. The input voltages shall be established at the minimum voltage, the mean voltage and the maximum voltage of the 18 to 30 Vdc operational voltage range. The inverter shall operate for a minimun of one hour at each of the input voltages with the minimum power factor loads required to obtain the factors of 0.8 leading and 0.6 lagging. The power factor shall be measured by means of a power factor meter.

4.6.9 Operability. - This test shall be conducted in accordance with the temperatures specified in 3.8. While stabilized at each of the specified temperature extremes specified in 3.8 the inverter shall be energized and continuously operated in accordance with the requirements specified in 3.8. Subsequent to the continuous operation at each temperature extreme, the inverter shall be subjected to the tests specified in 4.6.8 to 4.6.8.6 inclusive and shall conform to the requirements referenced therein.

4.6.10 Efficiency. - The inverter shall be inspected in accordance with the conditions specified in 3.9. Upon completion of the 10 minute stabilization period, the inverter shall be operated continuously as specified for 8 hours. Failure of the inverter to meet the requirements of 3.9 during continuous operation shall be cause for rejection.

4.6.11 Radio interference. - Inspection for radio interference shall be conducted in accordance with Test Methods and Procedures specified in Specification MIL-I-26600 to determine compliance with 3.10.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing and marking. - Packaging, packing and marking shall be in accordance with Specification MIL-P-14232 and Packaging Data Sheet MIL-P-14232/P10516412. The level of protection shall be as specified in the procurement document.

6. NOTES

6.1 Intended use. - This static inverter is primarily intended to supply ac power to the electronic and/or electrical components of various tanks. It is designed to operate from a 24 vdc source (storage batteries) and invert this voltage to 115 vac, 400 cps.

6.2 Ordering data. - Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Selection of applicable levels of preservation, packaging and packing.
(c) Applicable stock number.

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6.3 Qualification. - With respect to products requiring qualification, awards will be made only for such products as have, prior to the time set for opening of bids, been tested and approved for inclusion in the applicable Qualified Products List whether or not such products have actually been so listed by that date. The attention of the suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the Qualified Products List is U.S. Army Frankford Arsenal Phila., Pa. 19137 and information pertaining to qualification of products may be obtained from that activity.

6.4 Gradual temperature changes. - Gradual temperature changes are noted where the inverter is being prepared for the next temperature requirement. This is done for the protection of the inverter and may vary depending on how the inverter may be affected by the thermal shock.

Custodian:

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Preparing activity:

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Project No. 1220-A167

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